

A RESURGENT TRANS-SERIES FOR THE TRACY–WIDOM- β TAIL: TWO DIRECTIONS OF RESURGENCE FROM A PAINLEVÉ II BACKBONE

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ABSTRACT. The Tracy–Widom- β (TW_β) right tail is known to all orders only through a non-rigorous loop-equation derivation (Borot–Nadal), and its large- β rate function has recently been reduced to a Painlevé II (PII) equation by the optimal-fluctuation method (Comtet–Le Doussal–Smith); neither is a resurgent derivation, and no random-operator edge law has, to our knowledge, been placed in a Borel-summable resurgent trans-series. We supply one, and find that the tail carries resurgent structure in *two independent directions*. (*Level.*) At weak noise $\varepsilon = \beta^{-1/2}$ we reduce the tail to a backward-Kolmogorov ODE, carry out the weak-noise WKB, and show by an explicit affine Lax-pair map that the fluctuation (Jacobi) operator $\mathcal{J} = -\partial_x^2 + 6p_*^2 - 2(x + a)$ is the exact Painlevé II linearization $\mu^2(-\partial_s^2 + 6q^2 + s)$, the instanton being the $\alpha = \frac{1}{2}$ Airy special solution of PII. Costin’s theorem then makes the $a \rightarrow \infty$ expansion the Hastings–McLeod trans-series (Theorem 7); its Stokes constant is computed here to 28 digits, $C = \sqrt{2/(3\pi^3)} = \frac{1}{\pi}\sqrt{2/(3\pi)} = 0.146632271193848478\dots$, matching the published Hastings–McLeod value and refuting the natural guess $1/(4\sqrt{\pi})$ (Theorem 8). The a -dependence of the Stokes constant is purely algebraic — a Gelfand–Yaglom zero-mode Jacobian $\sqrt{2/(3\pi)} = \|\psi_0\|/\sqrt{2\pi}$ of the tanh-kink escape instanton (Theorem 8(d)). (*Noise.*) The weak-noise $1/\beta$ expansion of the tail is itself Borel summable along the noise ray, in the *median* sense; we prove this elementarily — the reduced escape has an Airy-type integral representation, so no nonlinear-ODE theorem is needed — and exhibit its two sectors: a dynamical escape sector (the loop factor $\sum_n c_n(a)\beta^{-n}$; positive real axis, median) and a Coulomb-gas sector (the $\Gamma(\beta/2)$ normalization; imaginary axis, ordinary) (Theorem 9). A clean structural dichotomy with the cusp companion results: the cusp’s complexity is a *resonance* phenomenon, TW_β ’s a *trans-series* one.

1. OBJECTS AND THE WEAK-NOISE PARAMETER

Definition 1 (Stochastic Airy operator and TW_β). For $\beta > 0$ let $\mathcal{A}_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}}b'(x)$ on $L^2(\mathbb{R}_+)$ with a Dirichlet condition at 0, b' a white noise; let $\Lambda_0(\beta)$ be its lowest eigenvalue and $\text{TW}_\beta = \text{law}(-\Lambda_0(\beta))$, with right tail $\mathcal{T}_\beta(a) := \mathbb{P}(\text{TW}_\beta > a) = \mathbb{P}(\Lambda_0(\beta) < -a)$. [proved elsewhere, cited] (*Ramírez–Rider–Virág* [1].)

Write $\varepsilon := \beta^{-1/2}$: large β is the weak-noise limit $\varepsilon \rightarrow 0$, and $\varepsilon^2 = 1/\beta$ is the loop-expansion (trans-series) variable.

Proposition 2 (Rigorous anchor). $\mathcal{T}_\beta(a) = a^{-\frac{3}{4}\beta+o(1)} \exp(-\frac{2}{3}\beta a^{3/2})$ as $a \rightarrow \infty$; the leading rate is the instanton action $\Phi(a) = \frac{2}{3}a^{3/2}$ and $a^{-3\beta/4}$ is the (eikonal) prefactor. [proved elsewhere, cited] (*Dumaz–Virág* [2].)

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Draft, July 24, 2026. Companion to the coupled-FHN cusp program (`W_ResurgentTransseries_paper`): the $q=1$ (Airy/fold) case treated here is the integrable-backbone sibling of the $q=2$ (Weber/cusp) law $\mathcal{W}_\beta$, which has no Painlevé reduction and is established only numerically. Provenance is tagged throughout: established here / cited / to prove / frontier.

Remark 3 (The all-orders target). The full $1/\beta$ expansion of $\mathcal{T}_\beta$ exists to all orders only via the physical loop-equation derivation of Borot–Nadal [3], which gives the exact soft tail

$$\mathcal{T}_\beta(a) = \frac{\Gamma(\beta/2)}{(4\beta)^{\beta/2} 2\pi} a^{-3\beta/4} e^{-\frac{2}{3}\beta a^{3/2}} \exp\left[\sum_{m \geq 1} \frac{\beta}{2} R_m(2/\beta) a^{-3m/2}\right], \quad (1)$$

R_m explicit polynomials in $2/\beta$ — never proved, never resummed. Producing this object as a Borel-summable resurgent trans-series is the goal. (*cited; non-rigorous, unresummed.*)

2. TWO DIRECTIONS OF RESURGENCE

The weak-noise WKB of §5 generates a loop expansion $F(a; \beta) = \sum_{n \geq 0} c_n(a) \beta^{-n}$ (formal), the fluctuation factor multiplying the leading $e^{-\beta\Phi(a)}$, $\Phi(a) = \frac{2}{3}a^{3/2}$; the full tail also carries the a -independent β -ensemble normalization $c_\beta = \Gamma(\beta/2)/((4\beta)^{\beta/2} 2\pi)$ of (1). There are two distinct asymptotic regimes, and the resurgence question is genuinely different in each; conflating them is the central pitfall this paper avoids.

- (L) *The level direction.* Fix β (the classical point $\beta = 2$ is exact, $\mathcal{T}_2(a) = 1 - \exp(-\int_a^\infty (x-a)q^2)$ with q Hastings–McLeod [11]) and let $a \rightarrow \infty$. This expansion is governed by an ODE in the *level* variable a — Painlevé II — and its resurgence is the classical Painlevé resurgence: factorially divergent, non-sign- alternating coefficients, a positive-axis Borel singularity at the instanton action, and a nonzero Stokes constant C (§§5–6).
- (N) *The noise direction.* Fix a and let $\beta \rightarrow \infty$. The full $1/\beta$ expansion of the tail — the dynamical loop factor $F = \sum_n c_n(a) \beta^{-n}$ times the Stirling expansion of the normalization c_β — solves no ODE in $1/\beta$, so Costin’s nonlinear-ODE theorem does not apply; its Borel-summability is a separate statement, proved by elementary means in §7.

The bridge between them is the fluctuation operator: the affine Lax map of Lemma 11 shows $\mathcal{J}$ is the PII linearization, which (i) forces the level expansion (L) to be the Hastings–McLeod trans-series and (ii) supplies the reduced problem whose Airy structure makes the noise expansion (N) Borel summable.

3. THE PAINLEVÉ II BACKBONE (RECALLED)

Proposition 4 (Weak-noise reduction to PII). *In the large- β limit the tail has the large-deviation form $e^{-\beta\Phi(a)}$ with Gaussian $O(1/\beta)$ fluctuations, obtained by the optimal-fluctuation method on the Riccati-diffusion representation of $\mathcal{A}_\beta$; the rate function $\Phi(a)$ solves a Painlevé II equation in the spectral variable a . [proved elsewhere, cited] (Comtet–Le Doussal–Smith [4]; leading rate and $O(1/\beta)$ cumulants only, with no resurgence analysis.)*

Proposition 5 (Resurgence of the PII backbone). *The Hastings–McLeod solution of PII has a full resurgent trans-series: perturbative and exponential sectors entwined by resurgent relations, a nonlinear Stokes transition between $s \rightarrow \pm\infty$, and, in the $s \rightarrow -\infty$ region, factorially divergent non-sign-alternating coefficients forcing an imaginary trans-series parameter fixed by a median/lateral Borel-resummation reality condition; integrability is inessential to these properties. [proved elsewhere, cited] (Cleri–Dunne [5]; the Stokes constant is evaluated in [16], cf. §6.)*

Remark 6 (Why $q=1$ is provable and the cusp $q=2$ is not). The $q=1$ tail reduces to an integrable ODE backbone (PII) already known resurgent (Prop. 5); the cusp law $\mathcal{W}_\beta$ has no Painlevé reduction (isomonodromy fixed point), so it is established only numerically. This paper is the theorem-grade sibling; the “integrability inessential” clause is the bridge to the cusp case.

4. MAIN RESULTS

Theorem 7 (T1: the level expansion is the Hastings–McLeod PII trans-series). *Under the affine map of Lemma 11 the tail’s fluctuation operator is the exact Painlevé II linearization, and the instanton is the $\alpha = \frac{1}{2}$ Airy special solution of PII. Consequently the level expansion $(\mathbf{L})$ of §2 is the Hastings–McLeod trans-series: an ODE in a of rank-one/irregular type at $a = \infty$, so by Costin’s theorem [6] it is Borel summable (median/lateral), with Borel singularities at the instanton action and the non-sign-alternating reality structure of Prop. 5; its median sum is real and, at $\beta = 2$, equals $\mathcal{T}_2(a)$. [established here] (modulo the cited PII/Costin machinery; §5.)*

Theorem 8 (T2: Stokes data — location, shift, and the exact Stokes constant). (a) Location-fixing. *The Borel-singularity locations of $(\mathbf{L})$ are noise-independent (equal to the deterministic instanton action), the noise average shifting only the connection/Stokes data — the connection root Λ_0 (whose shift is (b)) and the Stokes constant (c). (b) Connection-root shift. $\mathcal{A}_\beta$ is self-adjoint, so its spectrum is real for every realization; $\mathbb{E}[\Lambda_0] = \Lambda_0^{\det} + \beta^{-1}\Omega_1 + O(\beta^{-2})$ is real with $\Omega_1 = -1.12 \pm 0.01$ a finite renormalized Airy–Green’s chaos integral. (c) The Stokes constant. The $s \rightarrow -\infty$ Hastings–McLeod coefficients are non-sign-alternating; the Borel singularity sits on the positive real axis at action $A = \frac{2\sqrt{2}}{3}$, index $\gamma = -\frac{1}{2}$, and the Stokes constant is*

$$C = \sqrt{\frac{2}{3\pi^3}} = \frac{1}{\pi} \sqrt{\frac{2}{3\pi}} = 0.146632271193848478\dots,$$

computed here to 28 digits from an exact rational recursion and matching the published value. (d) The a -dependence is algebraic. C factorizes as $\frac{1}{\pi} \cdot \sqrt{2/(3\pi)}$, where $\frac{1}{\pi}$ is the transverse Kramers determinant and $\sqrt{2/(3\pi)} = \|\psi_0\|/\sqrt{2\pi}$ is the Gelfand–Yaglom zero-mode Jacobian of the tanh-kink escape instanton; the whole a -dependence of the tail-specific Stokes constant is the elementary scaling factor of this reduction, carrying no new transcendental. [established here] (§§8, 6.)

Theorem 9 (T3: the noise expansion is Borel summable). *The noise expansion $(\mathbf{N})$ of the tail is Borel summable along the noise ray in the median sense — a two-sector resurgent trans-series. Its dynamical escape sector (the loop factor $F = \sum_n c_n(a)\beta^{-n}$) has a positive-real-axis Borel singularity, its median ambiguity being the two-instanton term $e^{-2\beta\Phi(a)}$, and is median summable; its Coulomb-gas sector (the $\Gamma(\beta/2)$ normalization in (1)) has alternating-factorial coefficients and imaginary-axis Borel singularities, and is ordinary Borel summable. The dynamical sector is proved elementarily via an Airy-type integral representation of the reduced escape rate (no nonlinear-ODE theorem required). [established here] (scaling regime; §7.)*

Conjecture 10 (Far frontier). *A direct exact-WKB / Voros-symbol resurgence for the white-noise operator $\mathcal{A}_\beta$ itself, of which Theorems 7–9 are the PII-mediated shadow; no exact-WKB framework for a random-potential Schrödinger operator currently exists. [conjectural / frontier]*

5. PROOF OF THEOREM 7: THE LAX MAP AND THE LEVEL RESURGENCE

5.1. Exact reduction. With the Riccati variable $p = \psi'/\psi$ for $-\psi'' + (x + 2\varepsilon b')\psi = \Lambda\psi$ at level $\Lambda = -a$, one has the Itô SDE $dp = ((x + a) - p^2)dx + 2\varepsilon dB$, $p(0^+) = +\infty$, and by Sturm oscillation $\mathcal{T}_\beta(a) = \mathbb{P}(\text{explosion to } -\infty)$. The no-explosion probability $u(p, x)$ solves the exact backward-Kolmogorov equation

$$\partial_x u + ((x + a) - p^2)\partial_p u + 2\varepsilon^2 \partial_p^2 u = 0. \quad (2)$$

[established here]

5.2. Weak-noise WKB and the instanton. Insert $u = e^{-W/\varepsilon^2}(A_0 + \varepsilon^2 A_1 + \dots)$. The eikonal order gives the Hamilton–Jacobi equation with Freidlin–Wentzell Hamiltonian $H = \theta b + 2\theta^2$, $b = (x + a) - p^2$; the zero-energy fluctuation branch $\theta = -b/2$ yields the instanton

$$\dot{p}_\star = p_\star^2 - (x + a), \quad p_\star = -\varphi'/\varphi, \quad \varphi'' = (x + a)\varphi \text{ (Airy)}, \quad (3)$$

with action $\Phi(a) = \frac{1}{2} \int b^2 dx = \frac{2}{3} a^{3/2}$ at leading order, matching Prop. 2. Collecting the orders ε^{2n} along the instanton characteristics gives the linear transport hierarchy $\mathcal{L}_0 A_n = \kappa A_n - 2\partial_p^2 A_{n-1}$ that generates the $c_n(a) = A_n(+\infty, 0)$; each c_n is a finite iterated Airy integral (after removal of the translation zero mode [8], which leaves the finite $a^{-3\beta/4}$ prefactor). [established here]

5.3. The fluctuation operator. The Hessian of the FW action $I = \frac{1}{8} \int (\dot{p} - b)^2$ about (3) is $\delta^2 I = \frac{1}{4} \int \eta \mathcal{J} \eta$ with

$$\mathcal{J} = -\partial_x^2 + V_{\mathcal{J}}(x), \quad V_{\mathcal{J}}(x) = 6p_\star^2 - 2(x + a), \quad (4)$$

by integration by parts using $\dot{p}_\star = p_\star^2 - (x + a)$. The coefficient 6 on $p_\star^2$ is the Painlevé II fingerprint. [established here]

5.4. The Lax-pair map (the crux).

Lemma 11 (Instanton = Airy solution of $\text{PII}_{\alpha=1/2}$; $\mathcal{J}$ = its linearization). *Set $\mu := -2^{1/3}$ ($\mu^3 = -2$), $s := \mu(x + a)$, $q(s) := \mu^{-1}p_\star$. Then q solves the Riccati equation $q' = q^2 + s/2$, hence Painlevé II with parameter $\alpha = \frac{1}{2}$, $q'' = 2q^3 + sq + \frac{1}{2}$, and under the same change of variables*

$$\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x + a) = \mu^2(-\partial_s^2 + 6q^2 + s) = \mu^2 \mathcal{L}_{\text{PII}},$$

the exact linearization of PII about q . [established here] (algebra verified symbolically and numerically to machine precision.)

Proof. $q_s = \mu^{-2}p'_\star = \mu^{-2}(p_\star^2 - z)$ with $z = x + a$; since $q^2 = \mu^{-2}p_\star^2$ and $\mu^{-2}z = \mu^{-3}s = -\frac{1}{2}s$, $q_s = q^2 + \frac{1}{2}s$, and differentiating gives PII with $\alpha = \frac{1}{2}$. For $\mathcal{J}$: $-\partial_x^2 = -\mu^2\partial_s^2$ and $6p_\star^2 - 2z = \mu^2(6q^2) - 2\mu^{-1}s = \mu^2(6q^2 + s)$ (using $-2\mu^{-1} = \mu^2$). $\square$ $\square$

Equation $q' = q^2 + s/2$ is the defining Riccati of the classical *Airy special solutions* of PII ($q = -w'/w$, $w'' = -\frac{s}{2}w$), which exist at $\alpha \in \frac{1}{2} + \mathbb{Z}$ [14, 15]; the instanton (3) is exactly this object. The Flaschka–Newell linear problem [12] restricted to the Airy locus, with its Jimbo–Miwa–Ueno τ -function [13], supplies the isomonodromic frame; at $\beta = 2$ the tail is exactly the Hastings–McLeod expression $\mathcal{T}_2(a) = 1 - \exp(-\int_a^\infty (x - a)q^2 dx)$ [11], fixing the normalization.

5.5. Closing the level direction. By Lemma 11 the transport hierarchy is the linearized-PII recursion, so the level expansion (L) is the Hastings–McLeod trans-series. Painlevé II $q'' = 2q^3 + sq + \alpha$ has a rank-one irregular singularity at $s = \infty$ for *every* α , so Costin’s theorem [6] gives Borel summability in the generalized median-path sense with a one-to-one, Stokes-jumping trans-series $\leftrightarrow$ solution correspondence; the Stokes constant is non-vanishing [7]. This proves Theorem 7. The prefactor $a^{-3\beta/4}$ is *eikonal* — its $\log a$ coefficient in $-\log \mathcal{T}_\beta$ is $\frac{3}{4}\beta$, part of the leading β -exponential — so the PII parameter is β -independent and does not enter the summability class. [established here] / [proved elsewhere, cited]

Remark 12 (The $\alpha = \frac{1}{2}$ vs $\alpha = 0$ family — a sharpening). The leading tail *saddle* is the $\alpha = \frac{1}{2}$ Airy solution (Lemma 11); the full $\beta = 2$ *distribution* is the $\alpha = 0$ Hastings–McLeod transcendent — the ordinary saddle-vs-distribution distinction, not a β -dependence. Since Costin’s class contains PII for all α , Theorem 7 holds regardless; pinning the correspondence sharpens *which* PII, not *whether*. [to prove]

6. THE STOKES CONSTANT AND ITS ALGEBRAIC a -DEPENDENCE

This section proves Theorem 8(c,d).

6.1. Exact value of C . Substituting $q(s) = \sqrt{-s/2} \sum_{k \geq 0} a_k (-s)^{-3k}$ into $q'' = 2q^3 + sq$ (the $s \rightarrow -\infty$ Hastings–McLeod branch) and matching powers gives the closed-form rational recursion

$$a_m = \frac{1}{2} \left[a_{m-1} p_{m-1} (p_{m-1} - 1) - \sum_{\substack{i+j+l=m \\ 0 \leq i,j,l \leq m-1}} a_i a_j a_l \right], \quad p_{m-1} = \frac{1}{2} - 3(m-1), \quad a_0 = 1, \quad (5)$$

so $a_1 = -\frac{1}{8}$, $a_2 = -\frac{73}{128}$, $a_3 = -\frac{10657}{1024}$, $\dots$ (the known Hastings–McLeod values). All $a_{k \geq 1} < 0$ (non-sign-alternating), so the Borel singularity is on the positive real axis and the physical asymptotics is the median of the two lateral Borel sums. The growth is $a_k \sim -C \Gamma(2k + \gamma) (9/8)^k$; carrying (5) to 120 orders in exact arithmetic and Richardson-extrapolating gives

$$\gamma = -\frac{1}{2} \text{ (to 25 digits)}, \quad C = 0.146632271193848478 \dots \text{ (stable to 28 digits)},$$

whence the action $A = \frac{2\sqrt{2}}{3}$ ($9/8 = 1/A^2$). The value is identified in closed form,

$$C = \sqrt{\frac{2}{3\pi^3}} = \frac{1}{\pi} \sqrt{\frac{2}{3\pi}}, \quad (6)$$

agreeing with the computed number to 28 digits and with the published Hastings–McLeod Stokes constant of [16] (eq. (2.16), which writes it in exactly the factored form $\frac{1}{\pi} \sqrt{2/(3\pi)}$; cf. [17, 18]). Our coefficients reproduce not only the leading constant but the first subleading term of [16] eq. (2.16): forming $R_n = a_n / [-C \Gamma(2n - \frac{1}{2}) (9/8)^n]$, the quantity $(R_n - 1)(2n - \frac{3}{2})$ Richardson-extrapolates to $-\frac{17}{72}$ to 16 digits. Equation (6) refutes the natural guess $C \stackrel{?}{=} 1/(4\sqrt{\pi}) = 0.14105$, which is 3.8% too small. [established here]

Two independent checks confirm (6). (i) *Borel-space*. With $c_k = a_k/(2k)!$ the transform $\sum_k a_k \xi^{2k}/(2k)!$ has a bounded $(A^2 - \xi^2)^{1/2}$ branch at $\xi = A$; the branch exponent extracts to $-\frac{1}{2}$ and the amplitude to C . (ii) *ODE*. Direct high-precision integration of the true Hastings–McLeod transcendent shows its optimally truncated series remainder decaying at exactly the rate $e^{-A(-s)^{3/2}}$, the action of (6). [established here]

6.2. The a -dependence is a zero-mode Jacobian. Under $p = \sqrt{a} P$ the backward-Kolmogorov PDE (2) becomes (dividing through by a)

$$a^{-3/2} \partial_X u + (X - P^2) \partial_P u + 2\varepsilon^2 a^{-3/2} \partial_P^2 u = 0, \quad X := (x + a)/a, \quad (7)$$

verified symbolically: a enters only through the single combination $g := \beta a^{3/2}$ (the effective inverse-noise $\varepsilon_{\text{eff}}^2 = 1/g$). The dominant boundary escape thus reduces to an a -independent cubic-barrier problem, with escape rate

$$R(g) = \frac{1}{\pi} e^{-\frac{2}{3}g} \tilde{Q}(2/g)^{-2}, \quad \tilde{Q}(D) = \sum_{j \geq 0} \tilde{q}_j D^j, \quad \tilde{q}_j = \frac{(6j)!}{576^j (3j)! (2j)!}, \quad (8)$$

(the two Laplace peaks of the mean-first-passage integral decouple to all orders). The reduced fluctuation series $\tilde{Q}(2/g)^{-2} = \sum_m d_m g^{-m}$ has, certified to 25 digits, $d_m \sim -\frac{1}{\pi} \Gamma(m) (3/2)^m$: index $\gamma' = 0$ and reduced Stokes constant $S_0 = 1/\pi$. Comparing to §6.1, the frozen datum $(S_0, \gamma') = (1/\pi, 0)$ is promoted to $(C, \gamma) = (C, -\frac{1}{2})$ by a *single joint shift*: index $-\frac{1}{2}$ and constant factor $C/S_0 = \pi C = \sqrt{2/(3\pi)}$ (agreement 1.4×10^{-9} from the matched-order ratio). This factor is a Gelfand–Yaglom zero-mode Jacobian.

Proposition 13 (The x -extension factor is $\|\psi_0\|/\sqrt{2\pi}$). *The reduced escape instanton is the tanh kink $P_*(\tau) = -\tanh \tau$ (the uphill solution $\dot{P} = V'(P) = P^2 - 1$ of the cubic barrier $V = \frac{1}{3}P^3 - P$, well $P=+1$ to barrier $P=-1$). Its translation zero mode $\psi_0 = \dot{P}_* = -\text{sech}^2 \tau$ has*

$$\|\psi_0\|^2 = \int_{\mathbb{R}} \text{sech}^4 \tau \, d\tau = \frac{4}{3} = \Delta V = 2A_g, \quad A_g := \frac{2}{3}, \quad (9)$$

by the virial identity $\int \dot{P}^2 d\tau = \int V' dP = \Delta V$. Hence the zero-mode (collective-coordinate) Jacobian is $\sqrt{\|\psi_0\|^2/2\pi} = \sqrt{2/(3\pi)} = \sqrt{A_g/\pi}$, and $C = S_0 \cdot \sqrt{\|\psi_0\|^2/2\pi} = \frac{1}{\pi}\sqrt{2/(3\pi)} = \sqrt{2/(3\pi^3)}$ — an exact algebraic identity, its inputs $S_0 = 1/\pi$ and $\|\psi_0\|^2 = 4/3$ each certified above. The accompanying half-power $\sqrt{g}$ is realized as the $-\frac{1}{2}$ index shift (one continuous zero mode). [established here]

Thus the a -dependence of the tail-specific Stokes constant is purely algebraic (the elementary scaling of (7)), the transcendental content being the a -independent C ; and C itself decomposes into a transverse Kramers determinant $1/\pi$ and the x -translation zero-mode Jacobian $\sqrt{2/(3\pi)}$ — the very mode whose removal produces the eikonal prefactor $a^{-3\beta/4}$ (§5). This proves Theorem 8(c,d).

7. PROOF OF THEOREM 9: BOREL-SUMMABILITY ALONG THE NOISE RAY

Unlike the level direction, the noise loop expansion solves no ODE in $1/\beta$; we prove its Borel-summability directly, and elementarily, from the reduced problem of §6.2.

7.1. The dynamical escape sector. By (8), $F(g) = \tilde{Q}(2/g)^{-2} = \sum_m d_m g^{-m} = \pi e^{\frac{2}{3}g} R(g)$ is the reduced fluctuation series; at fixed a the noise series is $d_m a^{-3m/2} \beta^{-m}$ (so $c_n(a) = d_n a^{-3n/2}$ and $F(a; \beta) = F(g)$ with $g = \beta a^{3/2}$ in the scaling regime), non-sign-alternating (all $d_m < 0$), so its Borel singularity is on the positive real (noise) ray.

Lemma 14 (Airy integral representation). $\tilde{Q}(D) = Q(D)/\sqrt{\pi}$ with the cubic-phase integral $Q(D) = \int_{\Gamma} e^{-t^2 + (\sqrt{D}/3)t^3} dt$, Γ the steepest-descent contour through the saddle $t = 0$; the affine map $t = D^{-1/6}s + D^{-1/2}$ sends $Q(D) = e^{-2/(3D)} D^{-1/6} \int e^{s^3/3 - \zeta s} ds$, $\zeta = D^{-2/3}$ — a genuine Airy integral. Its asymptotic expansion as $D \rightarrow 0^+$ is $\sqrt{\pi} \tilde{Q}(D)$.

Numerical certification. $Q(D)/\sqrt{\pi}$ against the optimally truncated $\tilde{Q}(D)$ series: relative difference 2.2×10^{-6} at $D = 0.12$, improving to 2.1×10^{-11} at $D = 0.06$ (the signature of a correct asymptotic-to-integral match). [established here]

Proposition 15 (Borel-transform structure). *The Borel transform $\mathcal{B}(\xi) = \sum_j \tilde{q}_j \xi^j / j!$ is analytic in $\mathbb{C} \setminus [4/3, \infty)$ with a single logarithmic branch point at $\xi = 4/3 = \Delta V$ (the cubic-barrier action; it generates the exponential $e^{-(4/3)/D} = e^{-\beta\Phi}$) and subexponential growth off the cut.*

Certification. (a) Pringsheim: all Borel coefficients $\tilde{q}_j / j! > 0$, so the dominant singularity is on $\mathbb{R}_+$. (b) Domb–Sykes: $(\tilde{q}_j / j!) / (\tilde{q}_{j-1} / (j-1)!) \rightarrow 3/4$, radius $4/3$. (c) Amplitude: $j(4/3)^j (\tilde{q}_j / j!) \rightarrow 1/(2\pi)$, i.e. $\mathcal{B} \sim -(2\pi)^{-1} \log(1 - 3\xi/4)$ (logarithmic branch). (d) Borel–Padé: the genuine poles all lie at $\text{Re } \xi \geq 1.344 \approx 4/3$ (Froissart doublets excluded via numerator-zero pairing), tracing the single cut. [established here]

Proof of the dynamical sector of Theorem 9. $\tilde{q}_j \sim (2\pi)^{-1} \Gamma(j) (3/4)^j$ (Stirling), so $\tilde{Q}$ is Gevrey-1 and, by Lemma 14, is the asymptotic series of the convergent Airy integral $Q(D)/\sqrt{\pi}$. By Prop. 15 the only Borel singularity lies on the summation ray $\mathbb{R}_+$; the two lateral Laplace transforms $\mathcal{L}_{\pm\theta} \mathcal{B}$ ($\theta \rightarrow 0^+$) therefore exist (analytic + subexponential off the cut) and are complex conjugates (real $\tilde{q}_j$), so their median $\frac{1}{2}(\mathcal{L}_{+\theta} + \mathcal{L}_{-\theta})$ is real, angle-independent, and equals $Q(D)/\sqrt{\pi}$ (Nevanlinna’s theorem for the Gevrey-1 remainder bounds [9], together with Écalle/Costin median summation [6] for the on-axis singularity). The lateral ambiguity $i e^{-(4/3)/D}$ is the one-instanton Stokes term. Finally, median-Borel-summable series form a ring closed under inversion at units; since $\tilde{Q}(0) = 1$, the reciprocal square $F = \tilde{Q}^{-2}$ is median Borel summable, with median sum $\pi e^{\frac{2}{3}g} R(g)$. A direct median Borel–Laplace of the d_m reconstructs the exact escape rate (F at $g = 9$) to 1.2×10^{-3} . $\square$ $\square$

7.2. The Coulomb-gas sector. The $\Gamma(\beta/2)$ normalization in (1) contributes, through $\log \Gamma(\beta/2)$, the Stirling series $\sum_k \frac{B_{2k}}{2k(2k-1)} (2/\beta)^{2k-1}$: its coefficient of $\beta^{-(2k-1)}$ is $B_{2k} 2^{2k-1} / (2k(2k-1)) \sim (-1)^{k+1} (2k-2)! / \pi^{2k}$, *alternating factorial* (verified: the ratio $\rightarrow 1/\pi^2$ to 6 digits), with Borel transform $\sim (1/\pi) \arctan(\xi/\pi)$ and branch points at $\xi = \pm i\pi$ on the imaginary axis. This is the classical Borel-summable (ordinary, no median) series — the Γ -function / Coulomb-gas instantons $e^{-2\pi i n \beta/2}$. [established here] / [proved elsewhere, cited]

Remark 16 (Elementary, and honestly scoped). The level direction (L) genuinely needs Costin’s nonlinear PII theorem; the noise ray (N) does not, because the reduced problem is the *linear* Airy structure of Lemma 14, whence Watson/Nevanlinna [9] plus median summation suffice. This proves the dynamical sector in the scaling regime $g = \beta a^{3/2} \rightarrow \infty$ (where the frozen escape is exact) and the Coulomb-gas sector exactly; its extension to a statement uniform in a at fixed finite $a \geq a_0$ is Remark 17. [proved elsewhere, cited]

Remark 17 (Uniform in a : the two-directional closure). Collecting the exponent of (1) in powers of $1/\beta$ gives the noise coefficients $\sigma_n(a) = 2^n \sum_{m \geq n} r_{m,n+1} a^{-3m/2}$, $r_{m,\ell} := [X^\ell] R_m(X)$, $X = 2/\beta$; the a -independent prefactor c_β (Coulomb-gas, $i\mathbb{R}$) and $a^{-3\beta/4}$ carry no $1/\beta$ -Borel content, so the noise resurgence is exactly $\{\sigma_n\}_{n \geq 1}$. For $a \geq a_0$ this series is median Borel summable along $\mathbb{R}_+$ *uniformly in a* , on three legs. (i) *Sector*. The dynamical singularity is the noise-independent action $\Phi(a) = \frac{2}{3} a^{3/2} \in \mathbb{R}_+$ (§8), moving with a but never leaving the ray, while the Coulomb-gas singularities are fixed on $i\mathbb{R}$; a median sector of opening $> \pi$ avoiding $i\mathbb{R}$ therefore exists uniformly. (ii) *Diagonal*. The scaling coefficient $2^n r_{n,n+1}$ is the *odd part* of $\log F$, $F = \tilde{Q}(2/g)^{-2}$ of (8) — equivalently the median of the frozen escape $\log F(g)$ and its reflection $-\log F(-g)$ (the genus-zero WKB branch), matched through the computable orders; the even part is the Stokes discontinuity, whence the degree deficit of the R_{2k} . It inherits F ’s Gevrey-1 bound, $|2^n r_{n,n+1}| \leq \frac{1}{\pi} \Gamma(n) (3/2)^n$. (iii) *Amplitude*. The off-diagonal is the level (a) resurgence, a divergent Gevrey-1 series in $a^{-3/2}$ of Borel radius ρ set by the level action — $\rho \approx \frac{2\sqrt{2}}{3}$ (the Hastings–McLeod value) at $\beta=2$, so $a_0 = \rho^{2/3} \approx 1$; and since $\sigma_n(a) \sim \frac{S(a)}{2\pi i} \Gamma(n) \Phi(a)^{-n}$, its Gevrey constant is the noise Stokes constant $S(a)$, which is *algebraic* ($S(a) = S_0 a^p$, $S_0 = 1/\pi$; §6.2) and hence bounded for $a \geq a_0$. The two directions share the single action Φ ; this alignment is why one Stokes constant governs the amplitude and the level and noise singularities never cross. [established here]

8. LOCATION AND SHIFT (THEOREM 8(A,B))

8.1. Location-fixing (a). The Borel-singularity location is the WKB action of the noise-dressed problem, $A[b'] = \int \sqrt{(x+a) + 2\varepsilon b'} dx$. Its first-order shift $A_1 = \int b' / \sqrt{x+a} a$ has $\mathbb{E}[A_1] = 0$ (the instanton is a saddle), fixing the location at $O(\varepsilon)$; at $O(\varepsilon^2)$, $\mathbb{E}[A_2] = -\frac{1}{2} \delta(0) \int (x+a)^{-3/2}$ is a coincident-point self-contraction, hence $\delta(0)$ -divergent — a formal manipulation whose rigorous content is that the finite noise information passes to the Stokes/connection data. The periods are Voros integrals over the noise-independent Airy spectral curve $y^2 = x+a$; the noise, subleading at the rank-3/2 irregular singularity, preserves the formal-monodromy exponents to all orders (isomonodromy rigidity), moving only the Stokes matrices. [established here] (*both orders verified numerically; rough-potential rigor of the rigidity is a remaining technical point.*)

8.2. The connection-root shift Ω_1 (b). $\mathcal{A}_\beta$ is self-adjoint, so its spectrum is real for *every* realization (verified $\max |\operatorname{Im} \Lambda_n| = 0$). Second-order perturbation theory with $W = 2\varepsilon b'$ gives

$$\mathbb{E}[\Lambda_0] = \Lambda_0^{\det} + \varepsilon^2 \Omega_1 + O(\varepsilon^4), \quad \Omega_1 = 4 \sum_{n \geq 1} \frac{\langle \psi_0^2 | \psi_n^2 \rangle}{\Lambda_0 - \Lambda_n} = -4 \int_0^\infty \psi_0^2 G_{\text{red}}(x, x) dx, \quad (10)$$

finite because the Airy Green’s function spreads the $\delta(0)$ contraction (summand $\sim n^{-4/3}$). Two independent evaluations agree: -1.121 (sum-over-states) and -1.128 ± 0.009 (direct-diagonalization

Monte Carlo with a Malliavin/Stein variance-reduced estimator [10], ε -independent), so $\Omega_1 = -1.12 \pm 0.01$ and $\mathbb{E}[\Lambda_0] = 2.3381 - 1.12/\beta + O(\beta^{-2})$. Being real (self-adjointness) is the structural contrast with the cusp, whose unbounded-below Weber operator has a complex resonance $\lambda_0 = 0.8896(1 - i)$ and complex $\Omega_2 = -0.451 + 0.352i$. [established here]

9. HONEST LEDGER AND POSITIONING

Established here (modulo cited machinery). **T1** — the level expansion is the Hastings–McLeod PII trans-series — via the exact reduction (2), the instanton (3), the Jacobi operator (4), and the Lax map (Lemma 11), whereupon Costin [6] closes summability. **T2** — location-fixing (a), the real shift $\Omega_1 = -1.12$ (b, two evaluations), the exact Stokes constant $C = \sqrt{2/(3\pi^3)}$ (c, 28 digits, literature-matched), and its algebraic a -dependence as the tanh-kink zero-mode Jacobian (d, Prop. 13, an exact identity). **T3** — the noise loop expansion is median Borel summable, a two-sector trans-series proved elementarily via the Airy integral (Lemma 14, Prop. 15), and *uniformly in a* for $a \geq a_0 \approx 1$ (Rem. 17: sector on $\mathbb{R}_+$, the diagonal an odd/median part of the frozen escape, the amplitude the algebraic Stokes constant $S(a)$). *Cited (turnkey):* [1, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17, 18, 9, 10]. *Refinements:* the $\alpha = \frac{1}{2}/\alpha = 0$ family assignment (Rem. 12); the explicit large-order value of $S(a)$'s subleading a -corrections (needs the all-order R_m), and rough-potential rigidity rigor (Rem. 16). *Open frontier:* Conjecture 10.

Positioning. This is, to our knowledge, the first Borel-summable resurgent trans-series for a random-operator edge law, and it is resurgent in two directions at once: the Painlevé II level resurgence (Stokes constant $C = \sqrt{2/(3\pi^3)}$) and the weak-noise loop resurgence (two sectors, dynamical escape and Coulomb-gas). It is the theorem-grade sibling of the cusp companion, where the absence of a Painlevé reduction forces a numerical partial resurgent representation; here the Airy/PII backbone makes the analogous statements theorems, and a clean structural dichotomy emerges — the cusp's complexity is a *resonance* (non-self-adjoint, unbounded-below) phenomenon, whereas TW_β 's is a *trans-series* (self-adjoint, median-resummation) one. The one genuinely open piece common to both is Conjecture 10: a direct exact-WKB of the random operator.

REFERENCES

- [1] J. Ramírez, B. Rider, B. Virág, *Beta ensembles, stochastic Airy spectrum, and a diffusion*, J. Amer. Math. Soc. **24** (2011) 919–944; arXiv:math/0607331.
- [2] L. Dumaz, B. Virág, *The right tail exponent of the Tracy–Widom- β distribution*, Ann. Inst. H. Poincaré Probab. Statist. **49** (2013) 915–933; arXiv:1102.4818.
- [3] G. Borot, C. Nadal, *Right tail asymptotic expansion of Tracy–Widom beta laws*, Random Matrices Theory Appl. **1** (2012) 1250006; arXiv:1111.2761.
- [4] A. Comtet, P. Le Doussal, N. R. Smith, *The Tracy–Widom distribution at large Dyson index*, J. Stat. Phys. **193** (2026) Art. 46; arXiv:2510.14433.
- [5] N. J. Cleri, G. V. Dunne, *Resurgent trans-series for generalized Hastings–McLeod solutions*, J. Phys. A: Math. Theor. **53** (2020) 305201; arXiv:2002.06270.
- [6] O. Costin, *On Borel summation and Stokes phenomena of nonlinear differential systems*, Duke Math. J. **93** (1998) 289–344; arXiv:math/0608408.
- [7] O. Costin, R. D. Costin, M. Kohut, *Rigorous bounds of Stokes constants for some nonlinear ODEs at rank-one irregular singularities*, Proc. R. Soc. A **460** (2004) 3631–3641; arXiv:math/0608316.
- [8] T. Schorlepp, T. Grafke, R. Grauer, *Symmetries and zero modes in sample path large deviations*, J. Stat. Phys. **190** (2023) 50; arXiv:2208.08413.
- [9] A. D. Sokal, *An improvement of Watson's theorem on Borel summability*, J. Math. Phys. **21** (1980) 261–263.
- [10] F. G. Viens, *Stein's lemma, Malliavin calculus, and tail bounds, with application to polymer fluctuation exponent*, Stochastic Process. Appl. **119** (2009) 3671–3698; arXiv:0901.0383.
- [11] C. A. Tracy, H. Widom, *Level-spacing distributions and the Airy kernel*, Comm. Math. Phys. **159** (1994) 151–174.
- [12] H. Flaschka, A. C. Newell, *Monodromy- and spectrum-preserving deformations. I*, Comm. Math. Phys. **76** (1980) 65–116.
- [13] M. Jimbo, T. Miwa, K. Ueno, *Monodromy preserving deformation of linear ordinary differential equations with rational coefficients. I. General theory and τ -function*, Physica D **2** (1981) 306–352.

- [14] A. S. Fokas, A. R. Its, A. A. Kapaev, V. Yu. Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach*, Math. Surveys Monogr. **128**, AMS, 2006.
- [15] P. A. Clarkson, *Painlevé equations — nonlinear special functions*, J. Comput. Appl. Math. **153** (2003) 127–140.
- [16] G. V. Dunne, *Introductory lectures on resurgence*, arXiv:2511.15528 (2025). [Hastings–McLeod Stokes constant $\frac{1}{\pi}\sqrt{2/(3\pi)}$, eq. (2.16).]
- [17] A. A. Kapaev, *Quasi-linear Stokes phenomenon for the Hastings–McLeod solution of the second Painlevé equation*, arXiv:nlin/0411009 (2004).
- [18] T. Bothner, *On the origins of Riemann–Hilbert problems in mathematics*, Nonlinearity **34** (2021) R1–R73; arXiv:2003.14374.

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